



AR CAMPUS ASSISTANT

Campus navigation from outdoor routes to indoor facilities, schedules, shuttle context, and future AR guidance.

Problem

NUS students can reach a building but still lose time finding rooms, lifts, restrooms, printers, study spaces, and shuttle connections.

MILESTONE 1 COVERAGE

- 20 user stories across navigation, access, safety, transport, campus life, and community updates.
- 10 ideated features, including AR guidance, facility search, schedules, bus context, and map corrections.
- Project-level SWE: GitHub planning, tests, CI/CD, deployment, diagrams, and walkthrough evidence.

Core Features

- 1 **Auth**

email, demo, Google Sign-In, JWT
- 2 **Map**

MapLibre campus rendering
- 3 **Search**

place lookup and route tracing
- 4 **Assistant**

schedule-aware recommendations
- 5 **Facilities**

rooms, lifts, printers, study spaces
- 6 **Schedule**

create, view, delete items
- 7 **NUSMods**

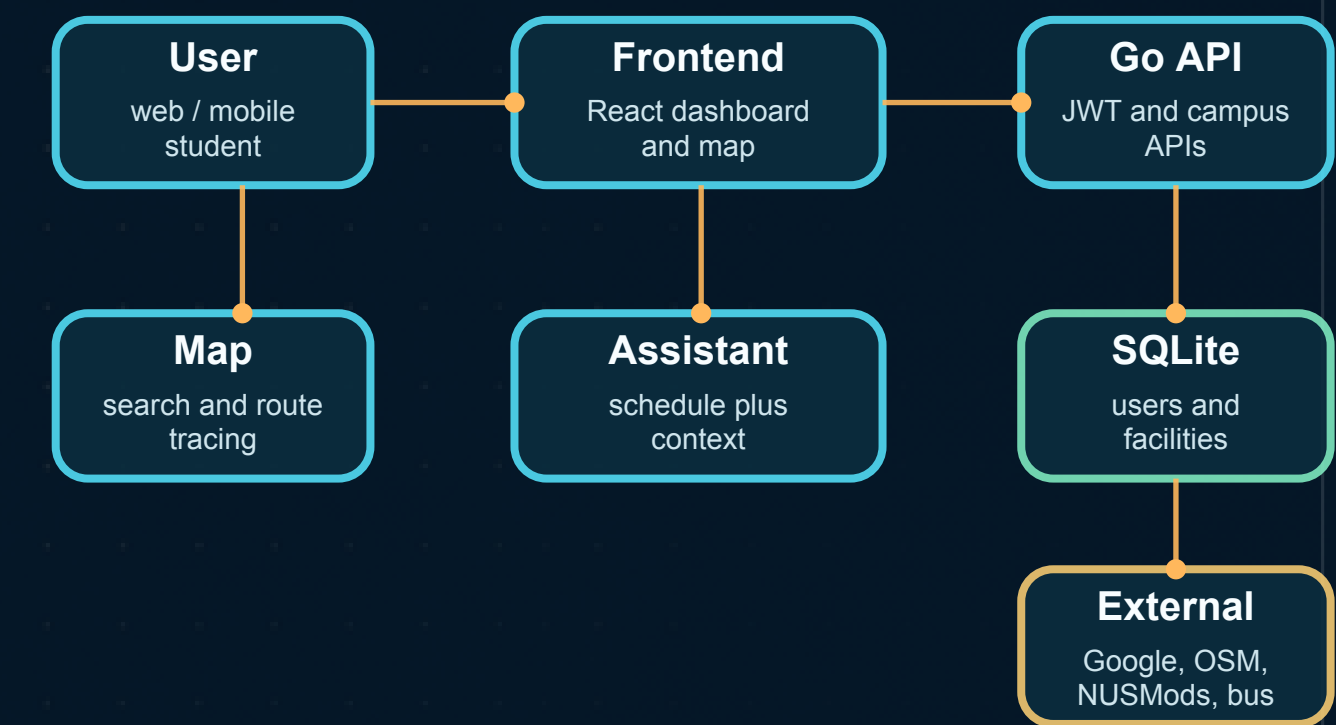
sync trigger and status
- 8 **Bus**

routes, stops, live arrivals
- 9 **Mobile**

responsive web and iOS shell
- 10 **SWE**

issues, tests, CI/CD, deploy

Architecture



Layered design separates UI, protected API, data, and external campus services.

Standard Diagrams

- 1 **Sequence**

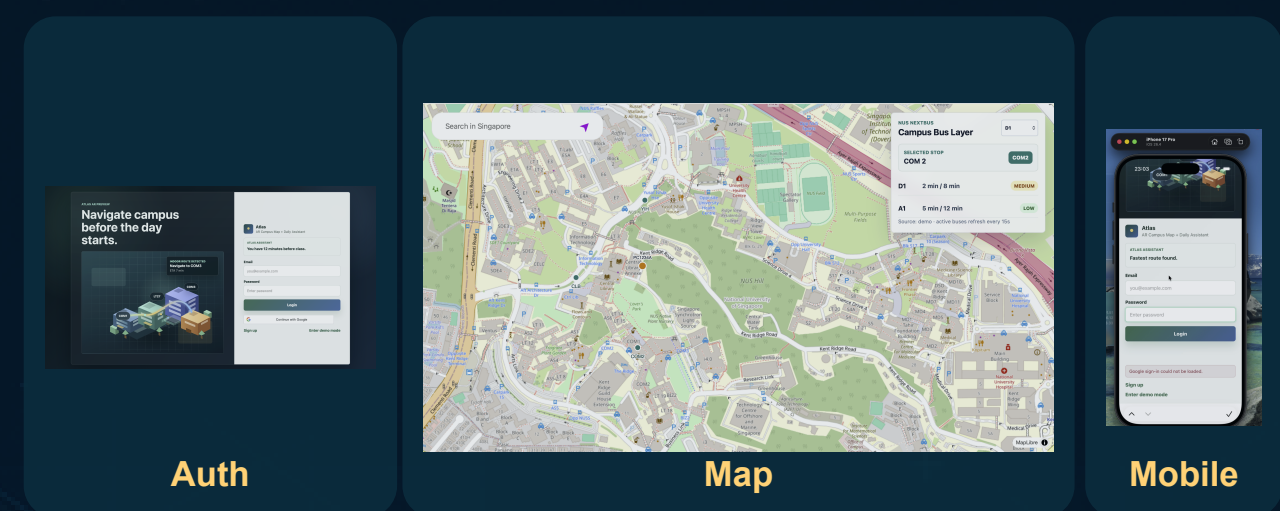
Login -> backend verification -> JWT -> protected dashboard.
- 2 **Activity**

Search location -> select place -> center map -> draw route.
- 3 **ER**

users, buildings, facilities, schedule_items, sync_status.
- 4 **Architecture**

frontend, backend, database, external campus services.

Prototype Screens



The walkthrough demonstrates login, protected dashboard access, schedule context, facility filtering, responsive layout, and campus map interaction.

Testing and CI/CD

- Backend**

cd backend && go test ./...
- Frontend**

cd frontend && npm run build
- iOS shell**

npm run ios:sync
- Deploy**

docker compose up -d --build

PIPELINE GATES

GitHub Actions plan: checkout, npm ci, frontend build, Go tests, Docker build, then deployment validation on Render.

User Story Coverage

- Navigation**

AR map, indoor guidance, travel time, saved locations.
- Campus Life**

study spaces, events, printing, less crowded facilities.
- Access**

wheelchair-friendly routes, lifts, accessible entrances.
- Transport**

nearby shuttle stops, routes, arrivals, bus context.
- Safety**

emergency exits, first-aid points, security offices.
- Community**

student corrections and admin review.

Walkthrough and Links

- 1 Open deployed app and login through demo/auth flow.
- 2 Show dashboard, schedule, recommendations, and facility filters.
- 3 Show map rendering, search interaction, and route tracing.
- 4 Explain repo logs, tests, CI/CD plan, and deployment setup.

App: orbital-artemis-armap-nus.onrender.com
GitHub: [77chenchen/Orbital_Artemis_ArMap_Nus](https://github.com/77chenchen/Orbital_Artemis_ArMap_Nus)